

Bhavani Adula

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SUMMARY

AI/ML engineer building and fine-tuning **generative AI** systems with **Hugging Face Transformers**, **Anthropic Claude**, and **RAG** — integrating LLMs with structured data via APIs and **PostgreSQL**-backed vector retrieval. Fine-tunes and evaluates transformer models in **PyTorch (BERT, T5)** with **SHAP** explainability and statistical validation, and engineers the data infrastructure behind them in **Python, SQL**, and **PostgreSQL** — from ingestion pipelines and hybrid **embedding + BM25** retrieval to reproducible **Git**-based workflows.

TECHNICAL SKILLS

Languages: Python, SQL, R
ML & Deep Learning: PyTorch, scikit-learn, XGBoost, BERT, T5, UMAP, HDBSCAN, SHAP, SciPy
GenAI & LLM: RAG, agentic workflows, tool calling, LLM orchestration, hybrid retrieval, vector search, safety guardrails, LLM evaluation, prompt engineering, Anthropic Claude API, Hugging Face Transformers, pgvector
Agent Reliability & Evaluation: LLM evaluation, eval datasets, regression testing, hallucination root-cause analysis, retrieval grounding, tool calling, function/argument validation, fallback and escalation logic, agent orchestration, p95 latency optimization
Data & Pipelines: PostgreSQL, Airflow, dbt, ETL/ELT, dimensional modeling, data validation, pandas, NumPy, Plotly, Tableau, Power BI, Excel
MLOps & Cloud: Docker, GitHub Actions, CI/CD, regression testing, observability, pytest, Streamlit, Git

EDUCATION

University of Missouri-Kansas City — MS, Computer Science | GPA 3.7 May 2026
J.B. Institute of Technology — B.Tech, Information Technology | GPA 3.2 May 2023

EXPERIENCE

Hippocratic AI — AI/ML Engineer Intern | USA (Remote) Apr 2025 - May 2026

- Built and owned **240+ LLM evaluation scenario cases** and eval datasets, analyzing and categorizing **300+ failed production conversations** through hallucination root-cause analysis; the regression suite caught **8 regressions pre-release** and cut repeat failures in the largest category **40%**.
- Hardened agent reliability across retrieval **grounding**, **tool-calling** argument validation, and **fallback/escalation logic** — raising document-grounded response accuracy **12%**, cutting invalid tool calls **70%** and missed escalations **45%**, and reducing **p95 latency 25%** with no quality regression.

Yrall Media Solutions — Data Engineer | India Jan 2023 - Jun 2024

- Built and maintained **Python/SQL ETL pipelines** on **PostgreSQL** and **Airflow** processing **20K-50K marketing records per day** from multiple sources, with data validation, incremental loading, and automated workflow monitoring.
- Modeled curated reporting datasets in **SQL/dbt** using **dimensional modeling**, supporting **200+ campaigns** and **Power BI** analytics across spend, clicks, conversions, **CTR**, cost per conversion, and **ROAS**.
- Made reported figures auditable with **record-level lineage** tracing each output to its source record, and wired a **data-quality framework into CI** to fail builds on regression.

PROJECTS

Silent Failure Detection — Label-Free ML Degradation Monitoring Aug 2026

- Built a domain-agnostic **drift-monitoring framework (8,209 lines Python, 261 tests)** backtesting a frozen credit model across **35 monthly windows** and **1,054,948 loans**, with **import-linter** contracts preventing the estimator from accessing the labels it is scored against.
- Showed **AUC held flat (0.672 to 0.666)** while **Brier rose 17%** and the **calibration gap widened 27x**; of six unsupervised detectors three fired on 100% of windows and one never fired — fixed-threshold alerting discards the trajectory signal.

ClinIQ — RAG Clinical Risk Detection | NSF NRT Research-A-Thon 2026, **4th Place** Apr - May 2026

- Owned temporal and behavioral analysis on a two-person team, clustering **3,316 SUD-flagged patient reviews** into **37 behavioral cohorts** with **UMAP** and **HDBSCAN** over 384-dim **all-MiniLM-L6-v2** embeddings.
- Tracked patient distress rising **1.7% to 30.5%** across **52,184 reviews (2008-2017)**, and authored the ethics and auditability framework grounding every **RAG** output in a cited government source.

Explainable Customer Churn Prediction Mar - May 2025

- Fine-tuned **BERT** for churn classification, reaching **0.793 accuracy** (macro F1 0.73) on **1,409 held-out customers**, and chained **T5** to convert structured customer rows into plain-language risk narratives.
- Ranked churn drivers with **SHAP** over **XGBoost** — contract type, tenure, and monthly charges — giving retention teams an auditable view they could act on without reading model scores.

CERTIFICATIONS & LEADERSHIP

Certifications: Google Data Analytics · Data Science with AI (upGrad) · Tableau for Data Science · Python (HackerRank)
Graduate Student Ambassador, UMKC School of Science and Engineering (Aug 2025 - Jul 2026) · **Membership Attendant**, UMKC Swinney Recreation Center, part-time (Jan 2026 - May 2026)

- Segmented a **~300-student international cohort** in **Excel**, surfacing enrollment trends that informed recruitment strategy; ran a student interview series and advised students.